Drugs and the large bowel

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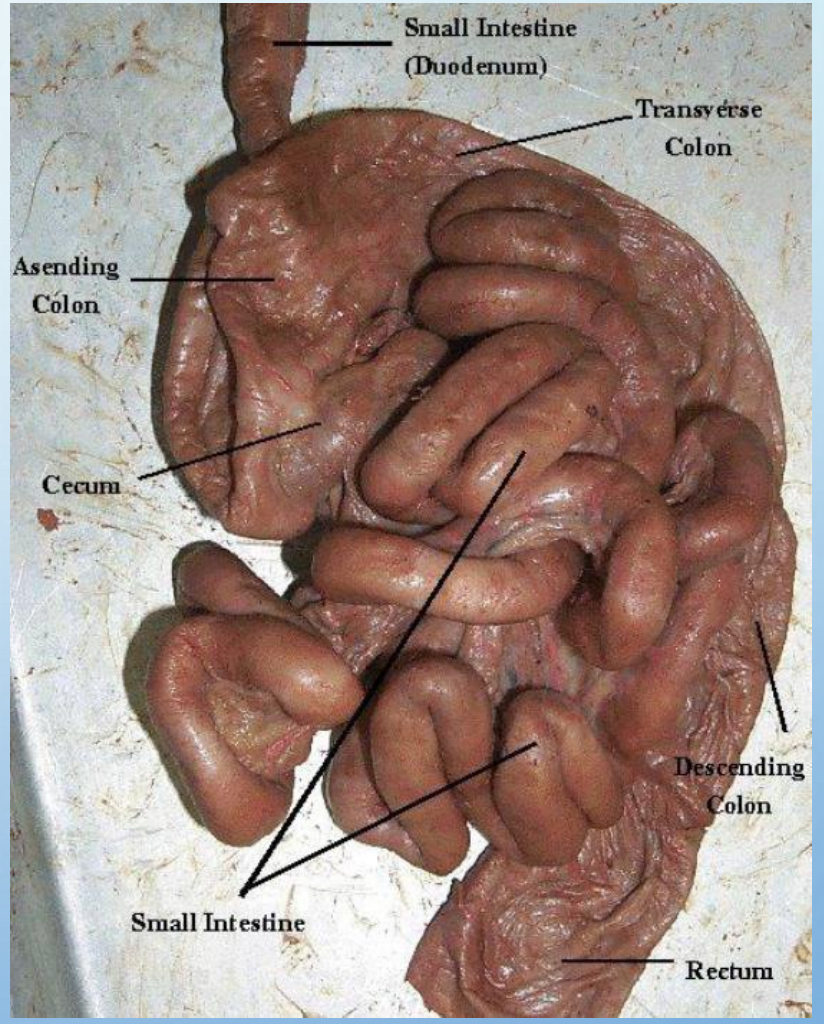
e-mail: jdn@hst.aau.dk

LEARNING OUTCOMES:

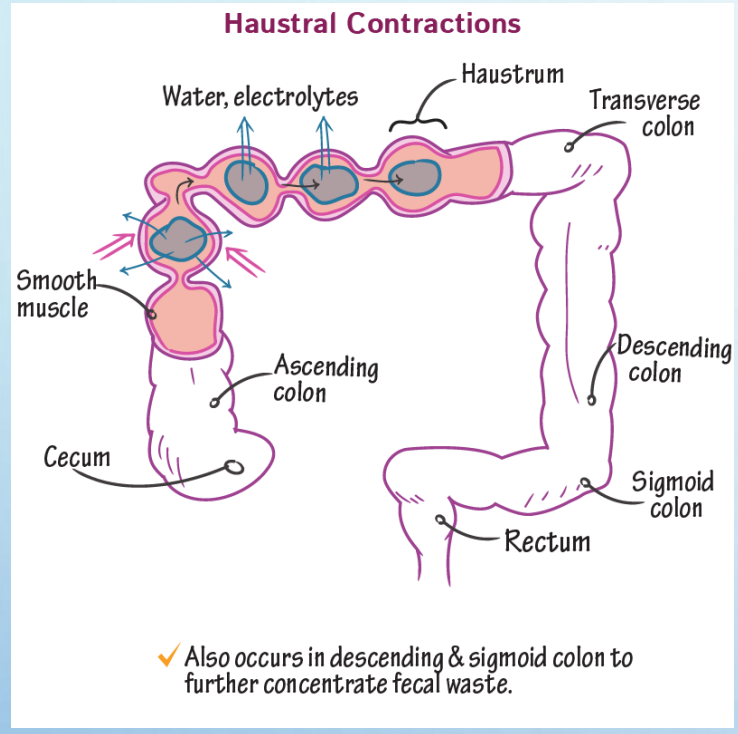
- Gøre rede for farmakologien for lægemidler, der virker på tarmen: anti-diarrémidler (obstipantia) og lægemidler, der bruges i forstoppelsesbehandling (laksantia)
- Forklar virkningen af opioidreceptor agonister på det enteriske nervesystem
- Beskriv konsekvenserne af overdreven brug af obstipantia og laksantia
- Forklar misbrug af laksantia ved spiseforstyrrelser
- Forstå den onde cirkel ved kronisk brug af laksantia

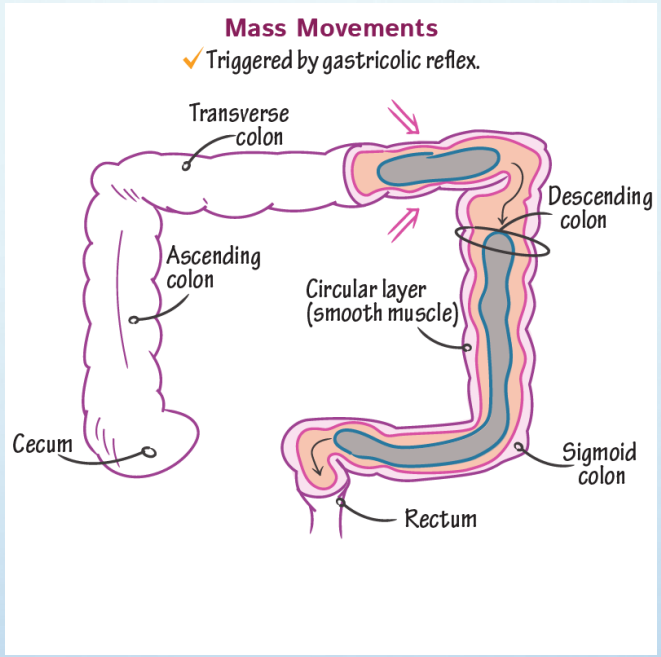
FUNCTION OF THE LARGE BOWEL

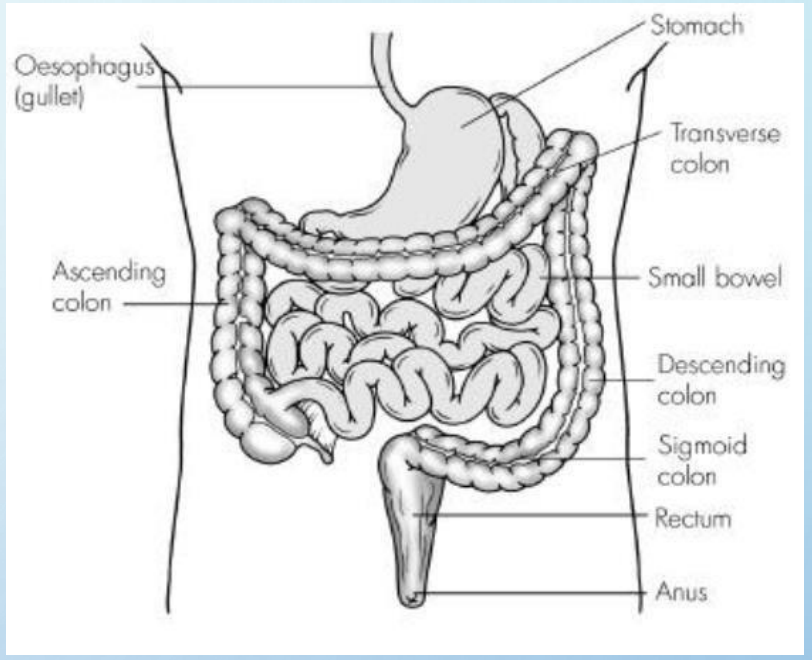
- Completing absorption
  (e.g. water, salts, amino acids, glucose, fatty acids);
  Primary function:
  volume reduction of chyme
- Regulation of colon mobility
- **Bacterial production of vitamins (K, B12, thiamine, riboflavin)**
- Storage & elimination of useless nutrients & wastes



COLON MOTILITY







- Proximal colon:
  haustration (mixing feces) approx. 2 contractions/hour
- Peristaltic wave: mass movement
  (propelling lumenal content forward and sweep colon clean 3-4 x/day

BRISTOL STOOL FORM SCALE

BRISTOL STOOL CHART

	Type 1	Separate hard lumps	Very constipated
	Type 2	Lumpy and sausage like	Slightly constipated
	Type 3	A sausage shape with cracks in the surface	Normal
	Type 4	Like a smooth, soft sausage or snake	Normal
	Type 5	Soft blobs with clear-cut edges	Lacking fibre
	Type 6	Mushy consistency with ragged edges	Inflammation
	Type 7	Liquid consistency with no solid pieces	Inflammation

BOWEL

- NORMAL FUNCTION
- DIARHEA
- OBSTIPATION

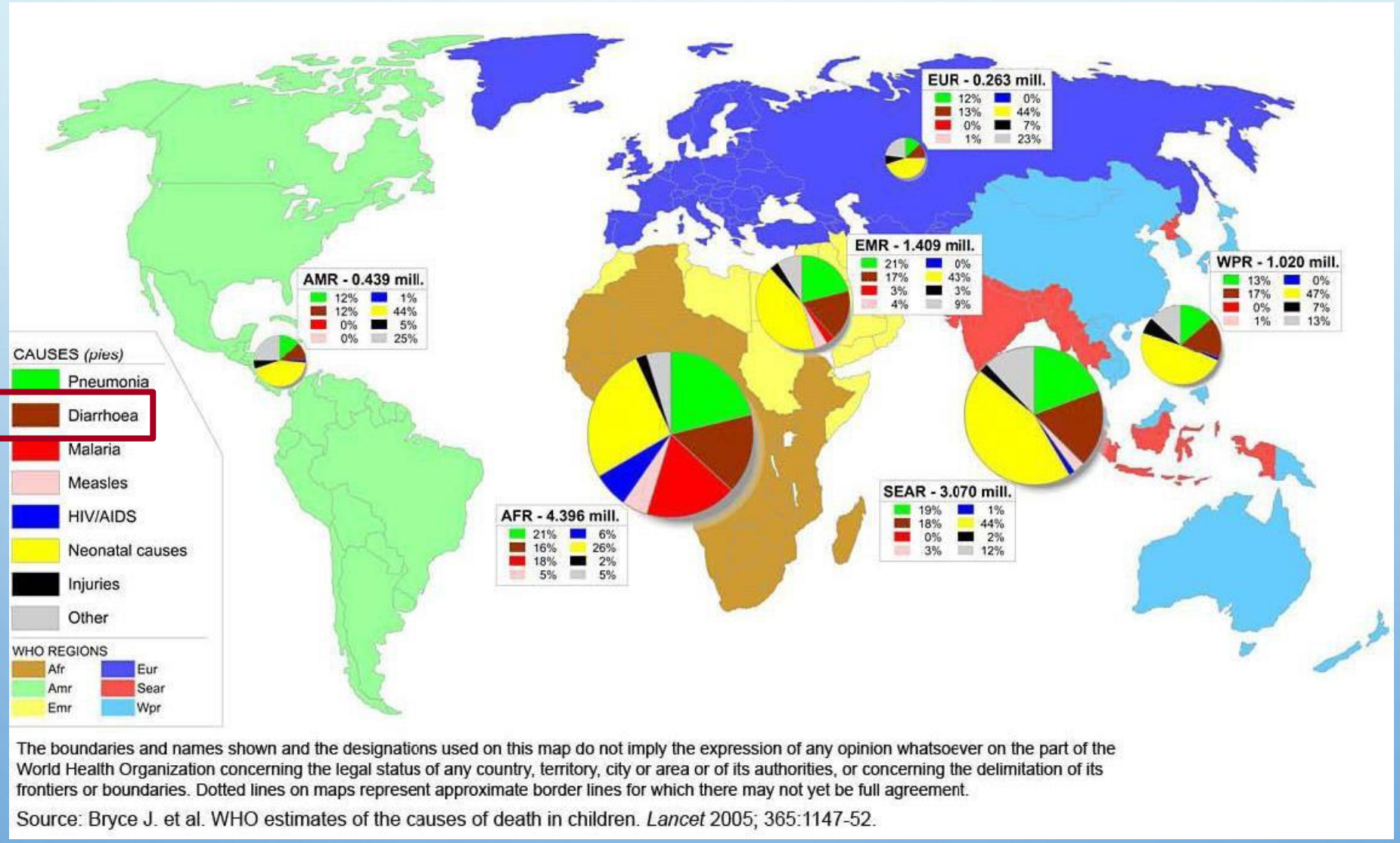
DIARRHEA

- ≥ 3 loose or liquid stools/d
- Increased frequency of bowel movements from 3-4 to ≥ 20 x/d
- Physiological imbalance of small & large intestine processes (ion absorption organic substrates, water)
- Causes: infection, malabsorption syndrome, enteropathies
- Most common complication: dehydration

- Ridding the gut of poisonous substances



<5 YRS MORTALITY (2000-2003)



The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted lines on maps represent approximate border lines for which there may not yet be full agreement.

Source: Bryce J. et al. WHO estimates of the causes of death in children. *Lancet* 2005; 365:1147-52.

DIARRHEA

- Increased GIT motility & secretion
- Decreased fluid absorption (loss of water & electrolytes)
- Bacterial toxins:
  increased electrolyte & fluid secretion via activation of GNRP (guanine nucleotide regulatory proteins) that couple surface receptors of mucosal cells to adenylate cyclase

Table **5** Medical assessment in diarrhea

Patient history	Physical examination
• Onset, stool frequency, type and volume • Presence of blood • Vomiting • Medicines received • Past medical history • Underlying conditions • Epidemiological clues	• Body weight • Temperature • Pulse/heart and respiratory rate • Blood pressure **Pediatric details**: Evidence of associated problems in children

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CAUSE OF DIARRHEA

Table 1 Overview of causative agents in diarrhea

Bacteria	Viruses	Parasites
• Diarrheagenic *Escherichia coli* • *Campylobacter jejuni* • *Vibrio cholerae O1* • *V. cholerae O139** • *Shigella species* • *V. parahaemolyticus* • *Bacteroides fragilis* • *C. coli* • *C. upsaliensis* • Nontyphoidal *Salmonellae* • *Clostridium difficile* • *Yersinia enterocolitica* • *Y. pseudotuberculosis*	• Rotavirus • Norovirus (calicivirus) • Adenovirus (serotype 40/41) • Astrovirus • Cytomegalovirus*	**Protozoan** • *Cryptosporidium parvum* • *Giardia intestinalis* • *Microsporida** • *Entamoeba histolytica* • *Isospora belli** • *Cyclospora cayetanensis* • *Dientamoeba fragilis* • *Blastocystis hominis* ***Helminths*** • *Strongyloides stercoralis* • *Angiostrongylus costaricensis* • *Schistosoma mansoni, S. japonicum*

* These agents are no longer reported in the Indian subcontinent.

Table **2** Episodes of diarrhea can be classified into three categories

Category	Clinical manifestation
Acute diarrhea	Presence of three or more abnormally loose or watery stools in the preceding 24 h
Dysentery	Presence of visible blood in stools
Persistent diarrhea	Acutely starting episode of diarrhea lasting more than 14 days

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DIAGNOSIS OF CULPRIT IN DIARRHEA

Table 4 Clinical features of infection with selected diarrheal pathogens

Pathogens	Clinical features					
	Abdominal pain	Fever	Fecal evidence of inflammation	Vomiting, nausea	Heme-positive stool	Bloody stool
Shigella	++	++	++	++	+/–	+
Salmonella	++	++	++	+	+/–	+
Campylobacter	++	++	++	+	+/–	+
Yersinia	++	++	+	+	+	+
Norovirus	++	+/–	–	++	–	–
Vibrio	+/–	+/–	+/–	+/–	+/–	+/–
Cyclospora	+/–	+/–	–	+	–	–
Cryptosporidium	+/–	+/–	+	+	–	–
Giardia	++	–	–	+	–	–
Entamoeba histolytica	+	+	+/–	+/–	++	+/–
Clostridium difficile	+	+	++	–	+	+
Shiga toxin-producing *Escherichia coli* (including O157:H7)	++	0	0	+	++	++

Key: ++, common: +, occurs, +/–, variable; –, not common: 0, atypical/often not present.

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ANTI-DIARRHEAL DRUGS

3 approaches to diarrheal treatment:

- Maintenance of fluid & electrolyte balance ORS (ORT)/Oral rehydration salt therapy: NaCl – 3.5 g/l
  KCl – 1.5 g/l
  Sodium citrate – 2.9 g/l
  Anhydrous glucose – 20.0 g/l
- Anti-infective agents
- Spasmolytic or other anti-diarrheal agents

ANTI-DIARRHEAL DRUGS

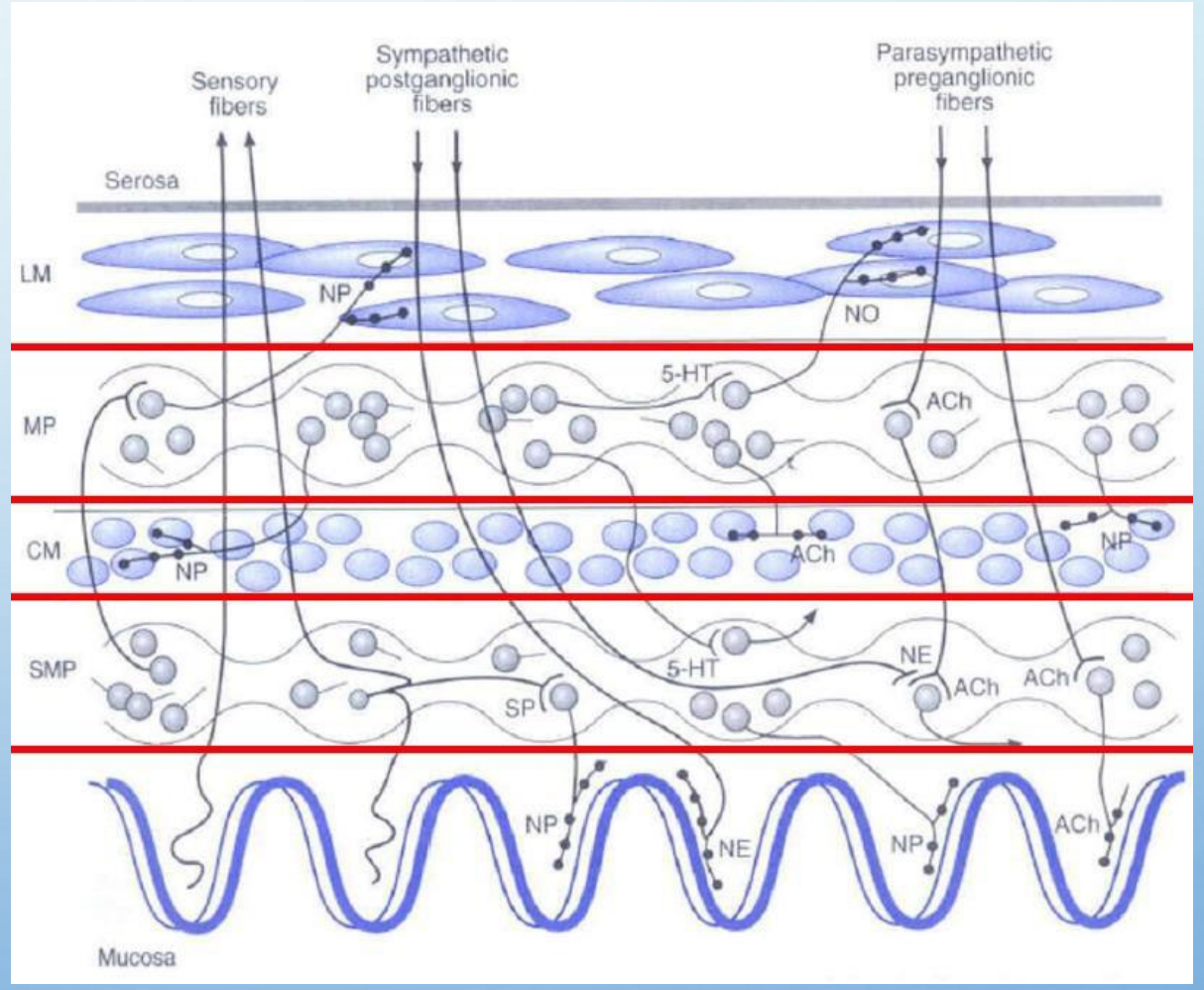
- ... may be safely used with mild to moderate acute diarrhea
- No applications in patients with bloody diarrhea, high fever, systemic toxicity: risk of worsening condition
- Discontinuation of drug administration if diarrhea is worsening despite therapy

ENTERIC NERVOUS SYSTEM

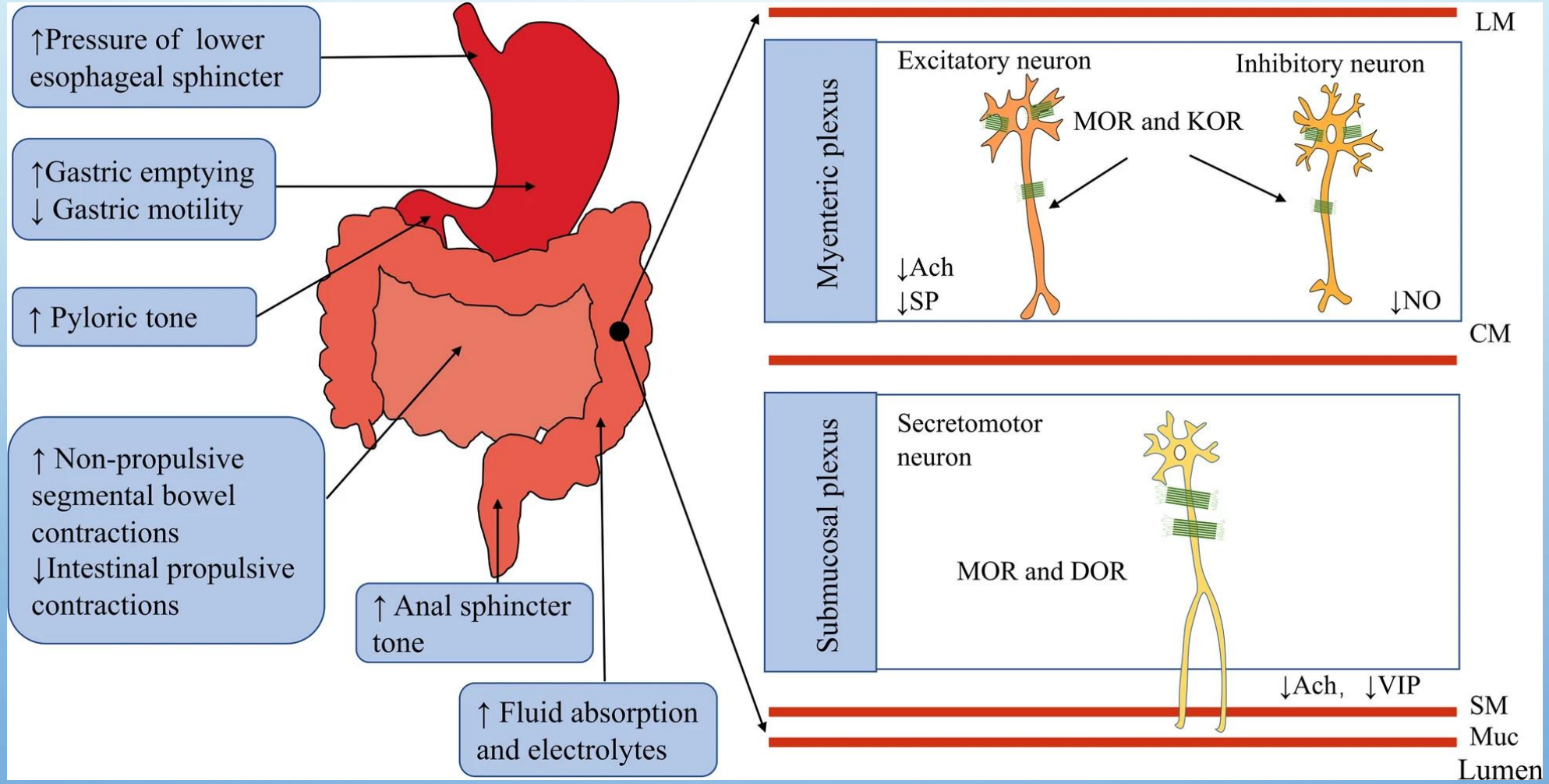
Intestine wall:

LM: longitudinal muscle layer
MP: myenteric plexus (Auerbach)
CM: circular muscle layer
SMP: submucosal plexus (Meissner)

NP: neuropeptides
NO: nitric oxide
5-HT: serotonin
Ach: acetlycholine
NE: norepinephrine
SP: substance P



Katzung, Basic & clinical pharmacology, 10th ed.


↑Pressure of lower esophageal sphincter
↑Gastric emptying
↓ Gastric motility
↑ Pyloric tone
↑ Non-propulsive segmental bowel contractions
↓Intestinal propulsive contractions
↑ Anal sphincter tone
↑ Fluid absorption and electrolytes
LM
Myenteric plexus
Excitatory neuron
Inhibitory neuron
MOR and KOR
↓Ach
↓SP
↓NO
CM
Submucosal plexus
Secretomotor neuron
MOR and DOR
↓Ach, ↓VIP
SM
Muc
Lumen

OPIOID AGONISTS

Mechanisms of action

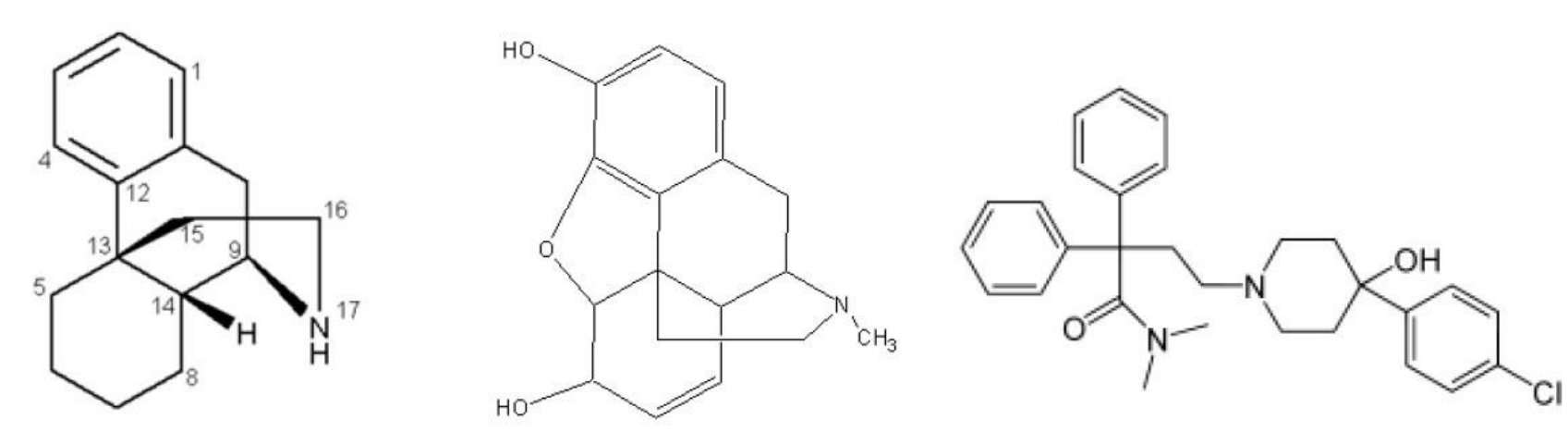
μ-opioid receptors: in myenteric plexus in large intestine; Inhibition of pre-synaptic cholinergic nerves in submucosal & myenteric plexuses

δ-opioid receptor: intestinal secretion (racecadotril)

- Decreased motility
- Diminished propulsive activity
- Pyloric, ileocolic and anal sphincters are contracted
- Decreased intestinal transit rate

Overall effect: constipation

μ OPIOID AGONISTS



morphinan

= base chemical opioid Structure

morphine

loperamide

- does not cross BBB:
  no central effects,
  no addictive potential
- no analgesic effect

OPIOID AGONISTS



Loperamide (e.g. Imodium®)

- Component of several anti-diarrheal drugs
- Relative selective action on GIT, undergoes significant enterohepatic cycling
- μ-opioid agonist
- Reduced frequency of abdominal cramps, decreased faecal passage, shortens duration of illness
- Contra-indication:
  diarrhea associated with organism that can penetrate the intestine walls (e.g. E.coli, Salmonella)

→ causes intestine immobilization
→ increased toxin production
→ reduced toxin elimination

OPIOID AGONISTS

Contra-indications/caution

- Some cases of severe infectious diarrhea (intereference with clearance from GIT):
  - Acute dysentery
  - Bacterial enterocolitis (*Salmonella*, *Shigella*, *Campylobacter*)
  - Pseudomembranous colitis secondary to broad spectrum antibiotics
- Severe inflammatory bowel disease with reduced GIT motility that could lead to ileus, megacolon, toxic megacolon



OPIOID AGONISTS

Diphenoxylate

- lacks morphine-like actvivity in CNS (standard dosage)
- prolonged use can lead to dependence
- μ-opioid antagonist
- preparation usually contain additionally atropine (muscarinic receptor antagonist) → anticholinergic effect contributes to anti-diarrheal action (reduced GI motility)

ADSORBENTS

Contain kaolin, pectin, calk, charcoal, methyl cellulose, activated attapulgite (Magnesium aluminium silicate)

- used in symptomatic treatment of diarrhea

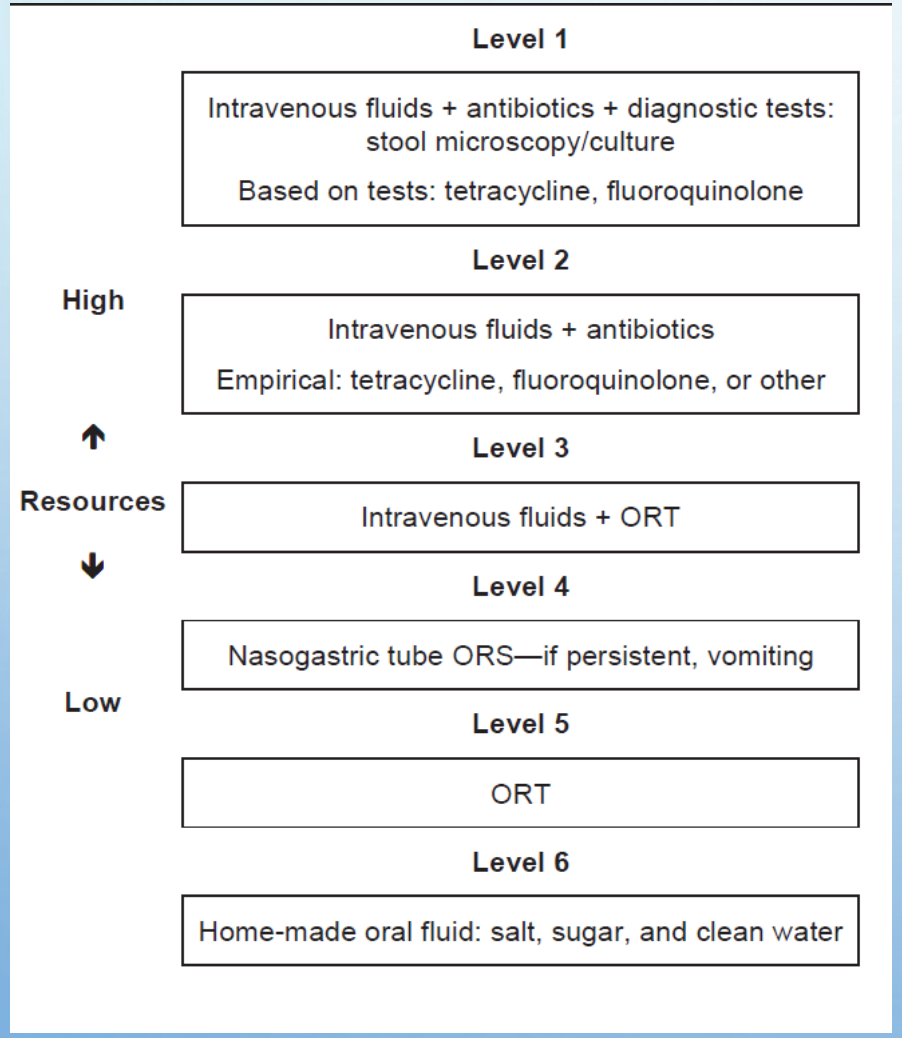
Mechanism of action:

- Adsorbing microorgansims or toxins
- Altering the intestinal flora
- Coating and protecting intestinal mucosa
  → decreasing stool liquidity and number

No hard evidence for this!
Adsorbents can be applied also to infants

CASCADES
HIERARCHICAL SET OF THERAPEUTIC OR DIAGNOSTIC TECHNIQUES

Acute, severe, watery diarrhea – severe dehydration



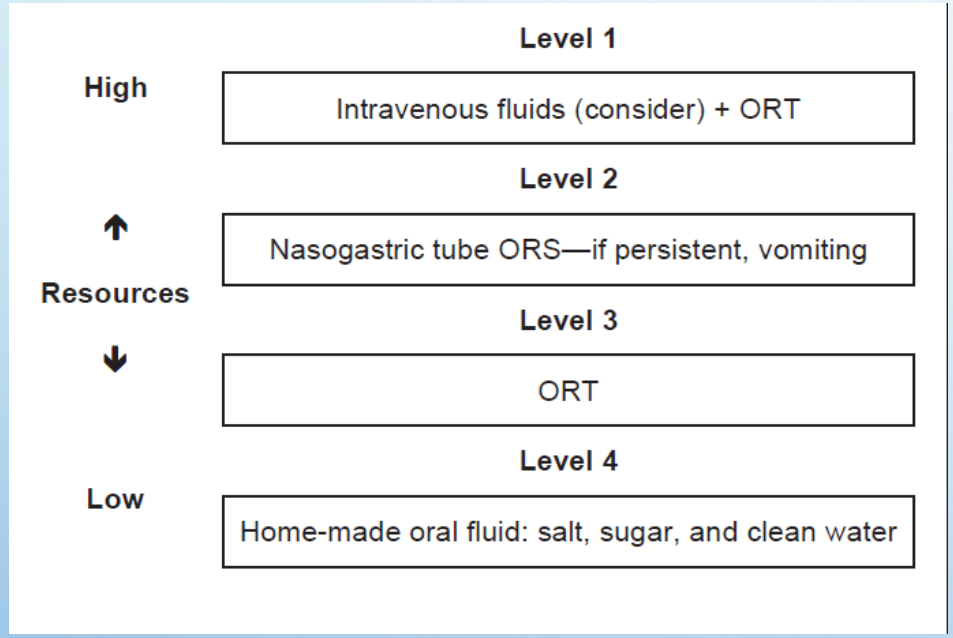
Home-made oral fluid:

- 1 level teaspoon of salt
- 8 level teaspoons of sugar
- 1 liter clean water (also preboiled)

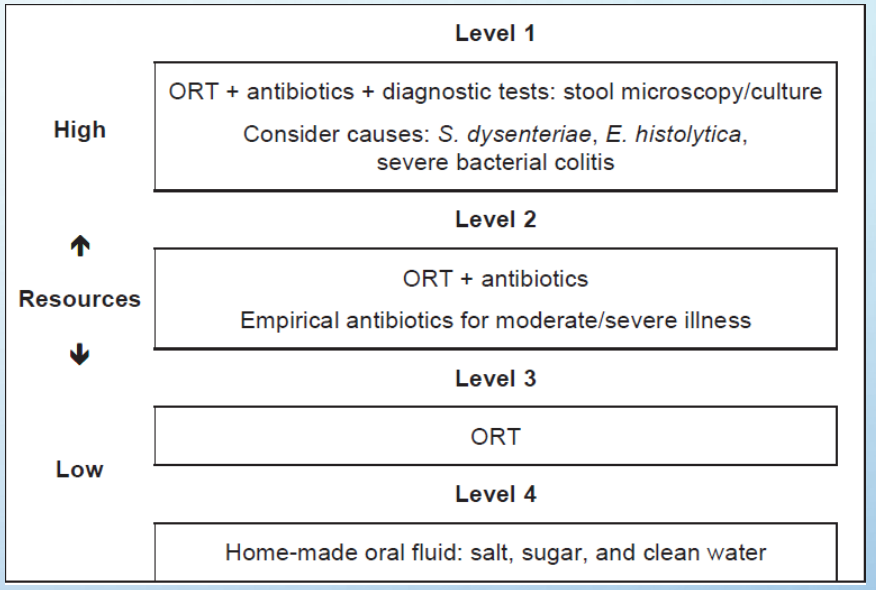
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CASCADES
HIERARCHICAL SET OF THERAPEUTIC OR DIAGNOSTIC TECHNIQUES

Acute, mild/moderate, watery diarrhea - mild/moderate dehydration



Acute, bloody diarrhea - mild/moderate dehydration



Home-made oral fluid:

- 1 level teaspoon of salt
- 8 level teaspoons of sugar
- 1 liter clean water (also preboiled)

World Gastroenterological Organization, 2012

CONSTIPATION



2 of the following:

- bowel movement <3x/w
- Stool usually hard, dry, small in size, difficult to eliminate
- Painful bowl movement, experience of straining, bloating, sensation of full bowel
- Sensation of incomplete evacuation
- Manual maneuvers
- Sensation of anorectal blockage/obstruction

CONSTIPATION



Risk factors:

- Aging
- Depression
- Inactivity
- Low calorie intake
- Low income & education level
- Number of medications taken
- Physical & sexual abuse

Table 1 Pathophysiology of functional constipation

Pathophysiologic subtype	Main feature, with absence of alarm symptoms or secondary causes
1 Slow-transit constipation (STC) • Colonic inertia • Colonic overactivity	Slow colonic transit of stool due to: • Decreased colonic activity • Increased, uncoordinated colon activity
2 Evacuation disorder	Colonic transit may be normal or prolonged, but evacuating stools from the rectum is inadequate/difficult
3 Constipation-predominant irritable bowel syndrome (IBS)	• Abdominal pain, bloating, altered bowel habit • May appear in combination with 1 or 2

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CONSTIPATION



Possible causes & constipation-associated conditions:

- Mechanical obstruction
  (e.g. colorectal tumor, megacolon)
- Neurological disorders/neuropathy (e.g.
  dementia, Parkinson)
- Endocrine/metabolic conditions
  (e.g. diabetes, dehydration)
- GI disorders & local painful conditions
  (e.g. irritable bowel syndrome, hemorrhoids)
- Myopathy
  (e.g. amyloidosis, systemic sclerosis)
- Dietary
  (e.g. low fiber, fluid depletion, anorexia)
- Miscallenous
  (e.g. cardiac diseases, degenerative joint diseases)

CONSTIPATION



Table 3 Medications associated with constipation

Prescription drugs
- Antidepressants
- Antiepileptics
- Antihistamines
- Antiparkinson drugs
- Antipsychotics
- Antispasmodics
- Calcium-channel blockers
- Diuretics
- Monoamine oxidase inhibitors
- Opiates
- Sympathomimetics
- Tricyclic antidepressants

Self-medication, over-the-counter drugs
- Antacids (containing aluminium, calcium)
- Antidiarrheal agents
- Calcium and iron supplements
- Nonsteroidal anti-inflammatory drugs

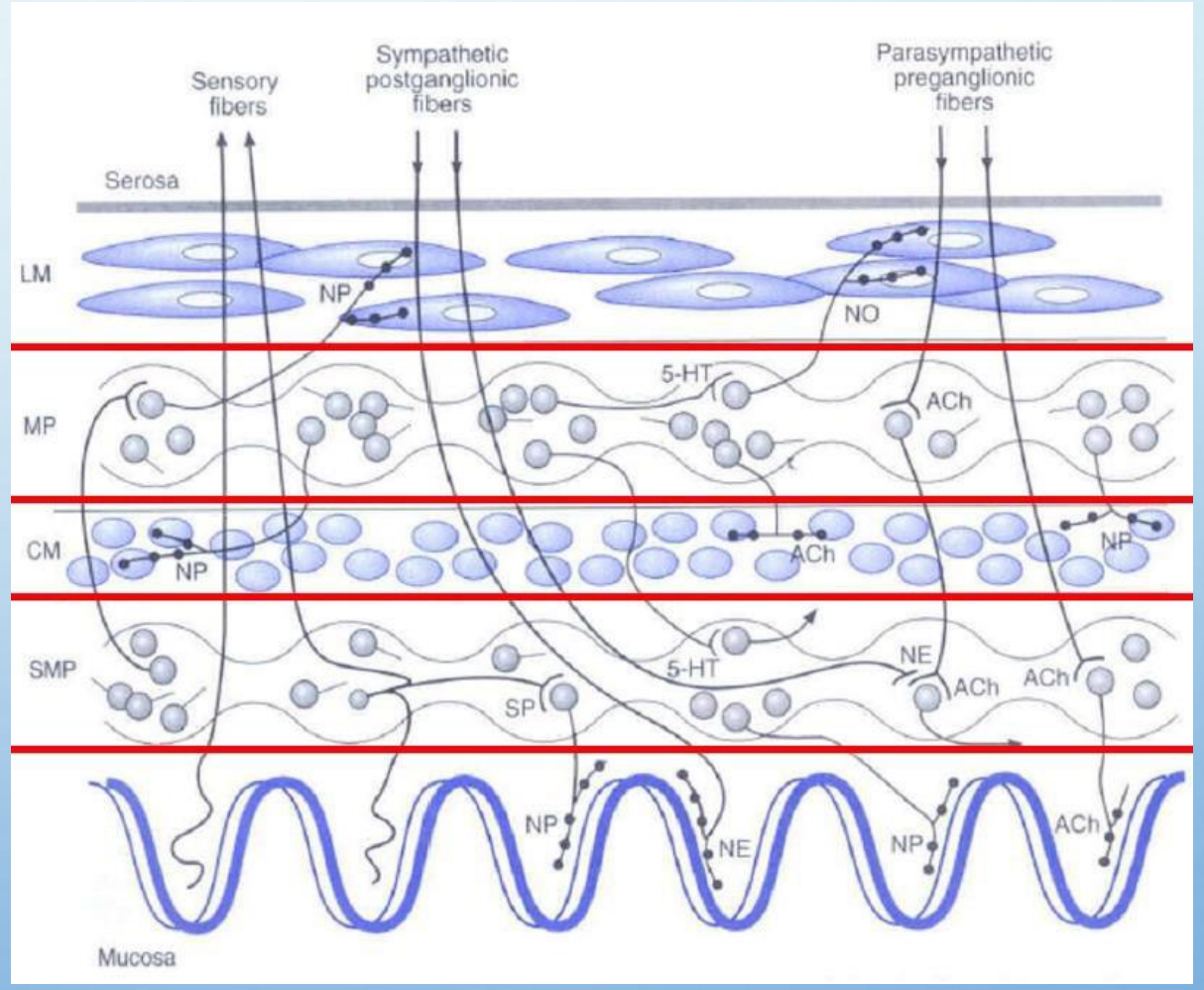
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ENTERIC NERVOUS SYSTEM

Intestine wall:

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NP: neuropeptides
NO: nitric oxide
5-HT: serotonin
Ach: acetlycholine
NE: norepinephrine
SP: substance P



Katzung, Basic & clinical pharmacology, 10th ed.

Constipation Indications for screening tests

Table 6 Physiologic tests for chronic constipation (reproduced with permission from Rao SS, Gastrointest Endosc Clin N Am 2009;19:117–39)

Test	Strength	Weakness	Comment
Colonic transit study with radiopaque markers	Evaluates the presence of slow, normal, or rapid colonic transit; inexpensive and widely available	Inconsistent methodology; validity has been questioned	Useful for classifying patients according to pathophysiological subtypes
Anorectal manometry	Identifies evacuation disorder, rectal hyposensitivity, rectal hypersensitivity, impaired compliance, Hirschsprung disease	Lack of standardization	Useful for establishing diagnoses of Hirschsprung disease, evacuation disorder, and rectal hyposensitivity or hypersensitivity
Balloon expulsion test	Simple, inexpensive, bedside assessment of the ability to expel a simulated stool; identifies evacuation disorder	Lack of standardization	Normal balloon expulsion test does not exclude dyssynergia; should be interpreted alongside results of other anorectal tests

Laboratory studies, function tests, imaging or endoscopy indicated in:

- Patients with severe constipation or alarm symptoms

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Constipation Classification

Table 7 Constipation categories based on clinical evaluation

Constipation type	Typical findings
Normal-transit constipation, constipation-predominant IBS	• Patient history, no pathology at physical inspection/examination • Pain and bloating • Feeling of incomplete evacuation
Slow-transit constipation	• Slow colonic transit • Normal pelvic floor function
Evacuation disorder	• Prolonged/excessive straining • Difficult defecation even with soft stools • Patient applies perineal/vaginal pressure to defecate • Manual maneuvers to aid defecation • High basal sphincter pressure (anorectal manometry)
Idiopathic/organic/secondary constipation	• Known drug side effects, contributing medication • Proven mechanical obstruction • Metabolic disorders—abnormal blood tests

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Constipation Classification

Table 8 General management of constipation

1. Patient history + physical examination	
↓	
2. Classify the patient's type of constipation—see Table 7 (constipation categories)	
↓	
3. Medical approach in uncomplicated normal-transit constipation without alarm symptoms	• Fiber, milk of magnesia • Add lactulose/PEG • Add bisacodyl/sodium picosulfate • Adjust medication as needed
↓	
4. In treatment-resistant constipation, specialized investigations can often identify a cause and guide treatment	• Standard blood test and colonic anatomic evaluation to rule out organic causes; manage the underlying constipation causing the pathology • The majority of patients will have a normal/negative clinical evaluation and may meet the criteria for constipation-predominant IBS. These patients will probably benefit from treatment with fiber and/or osmotic laxatives
↓	
5. If treatment fails, continue with specialized testing (this may only apply to the "extensive resources" level)	• Identify STC with a radiopaque marker study • Exclude evacuation disorder with anorectal manometry and balloon expulsion test • Evaluate anatomic defects with defecography
↓	
6. Treatment of STC with aggressive laxative programs	• Fiber, milk of magnesia, bisacodyl/sodium picosulfate • Prucalopride, lubiprostone • Add lactulose/PEG if no improvement • In refractory constipation, a few highly selected patients may benefit from surgery

IBS, irritable bowel syndrome; PEG, polyethylene glycol; STC, slow-transit constipation.

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LAXATIVES

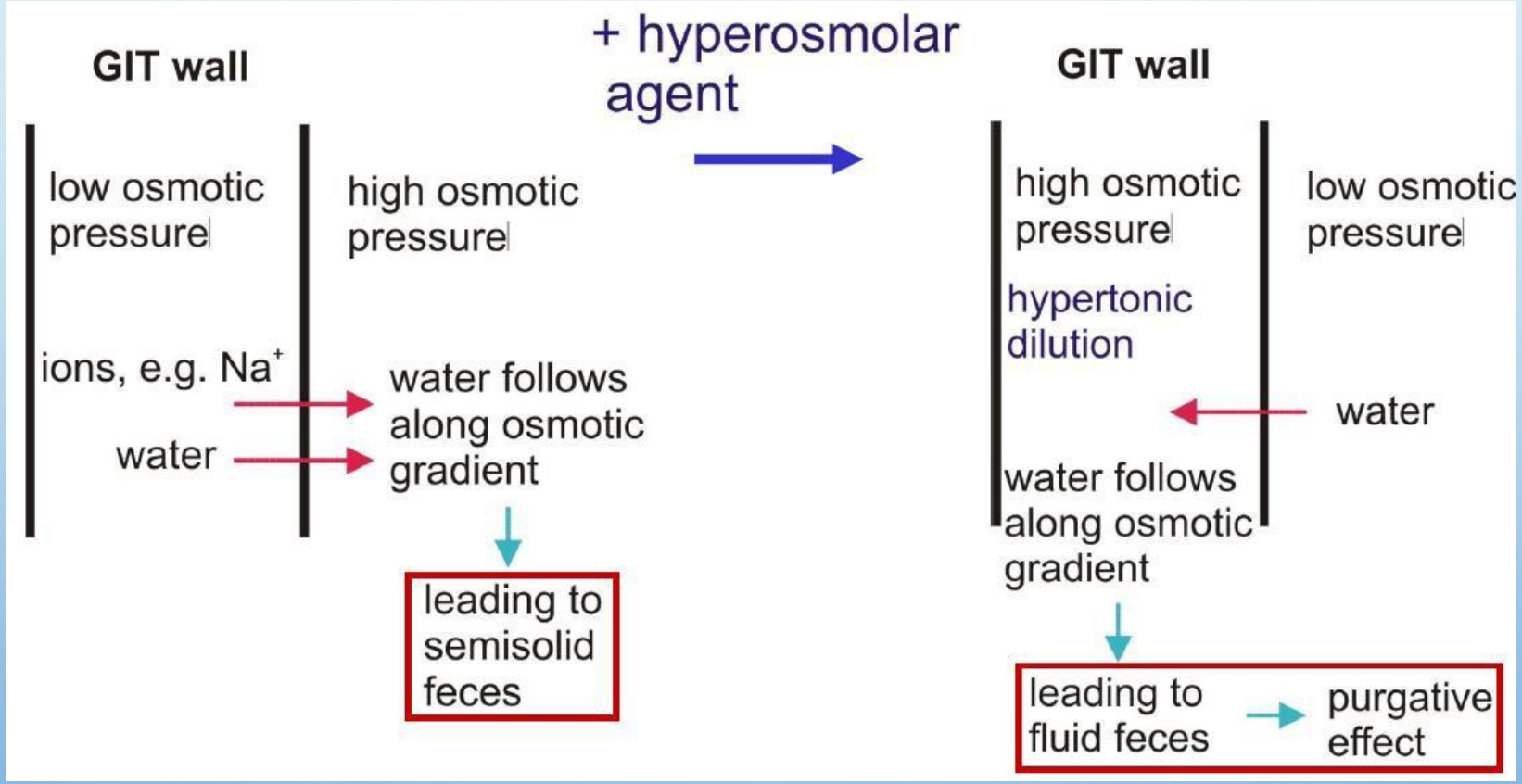
- Bulk laxatives
- Osmotic laxatives
- Stool softeners
- Stimulant laxative
- Serotonin 5-HT_4-receptor agonist

BULK-FORMING LAXATIVES

Methylcellulose, plant extracts (sterculia), psyllium, agar

- Polysaccharide polymers that are not broken down during digestion in the upper GIT
  → form bulky hydrated mass in the gut lumen promoting peristalsis and improving faecal consistency
- May take several days to work
- No side-effects

OSMOTIC LAXATIVES



OSMOTIC LAXATIVES

Magnesium sulfate, citrate, sodium phosphate

- Producing osmotic load
  → trap increased volumes of fluid lumen in bowel
  → accelerating transfer of gut contents through intestine
  → abnormally large volume entering the colon
  → causes distension & purgation within 1 h.

- May take lead to intravascular volume depletion and electrolyte fluctuations
  → can cause heart block, neuromuscular block, CNS depression
  → should be avoided in small children & patients with poor renal function

OSMOTIC LAXATIVES

Lactulose, sorbitol (non-absorbable sugar)

- Chronic constipation (e.g. morphine side-effects)
  → can be taken on daily basis for longer time periods (decades, unlike most other laxatives)
- Draw water in bowel → casue cathartic effect through osmotic action

BUT:

- Metabolized by colonic bacteria
  → side-effects include severe flatus, cramps
- Can take 1-2 d to work, generally produces results within 4 h if taken on daily basis

OSMOTIC LAXATIVES

Polyethylene glycol (PEG)

- Isotonic solution, non-absorbable, osmotically active
  → binds water
  → transports water to colon
  → increase osmotic pressure (locally)
  → release of water into the colon
- No significant intra vascular fluid or electrolyte shifts
  → safe for all patients
- Complete colonic cleaning prior to GIT endoscopic procedure

STOOL SOFTENERS

- Docusate sodium, glycerin, arachis oil, liquid paraffin
  - Surface-active compound
  - Act in GIT in a manner similar to a detergent and produces softer
    - Faeces, permit water and lipids to penetrate
  - Also weak stimulant laxative
  - Docusate prescribed to prevent constipation in hospitalized
    - Patients
  - Glycerin suppository prescription for infants

STIMULANT LAXATIVES

Mechanism of action:

- Increased peristalsis possibly by stimulating entercic nerves
- Act mainly by increasing electrolyte
  → hence, water secretion by mucosa
- Abdominal cramping may be experience as a side-effect
  with almost any of these drugs

STIMULANT LAXATIVES

Cascara, aloe, senna

- Occur naturally in plants, anthroquinone laxatives
- Bowel movement in 6-12 h after oral administration, within 2 h when given rectally
- Active principle (after hydrolysis of glycosidic linkages): directly stimulates the myenteric plexus
  → increased peristalsis
  → defecation

STIMULANT LAXATIVES

Bisacodyl (Dulcolax®), sodium picosulfate (Laxoberal®)

- Diphenylmethane derivates
- Use of sodium picosulfate associated with certain electrolyte disturbances such as hyponatremia and hypokalemia
- Long-term medication >1-2 w can lead to intestinal hypomotility

SEROTONIN 5-HT_4-RECEPTOR AGONIST

- Tegaserod
  - High affinity for 5-ht_4-receptors but not appreciable binding to 5-ht_3- or dopamine receptors
  - Stimulation of 5-ht_4-receptors
    - → Release of neurotransmitters
    - → Promote peristaltic reflexes
  - Promotes gastric emptying, enhances small and large bowel transit
  - Activates camp-dependent chloride secretion from colon
    - → Increased stool liquidity

LAXATIVE ABUSE

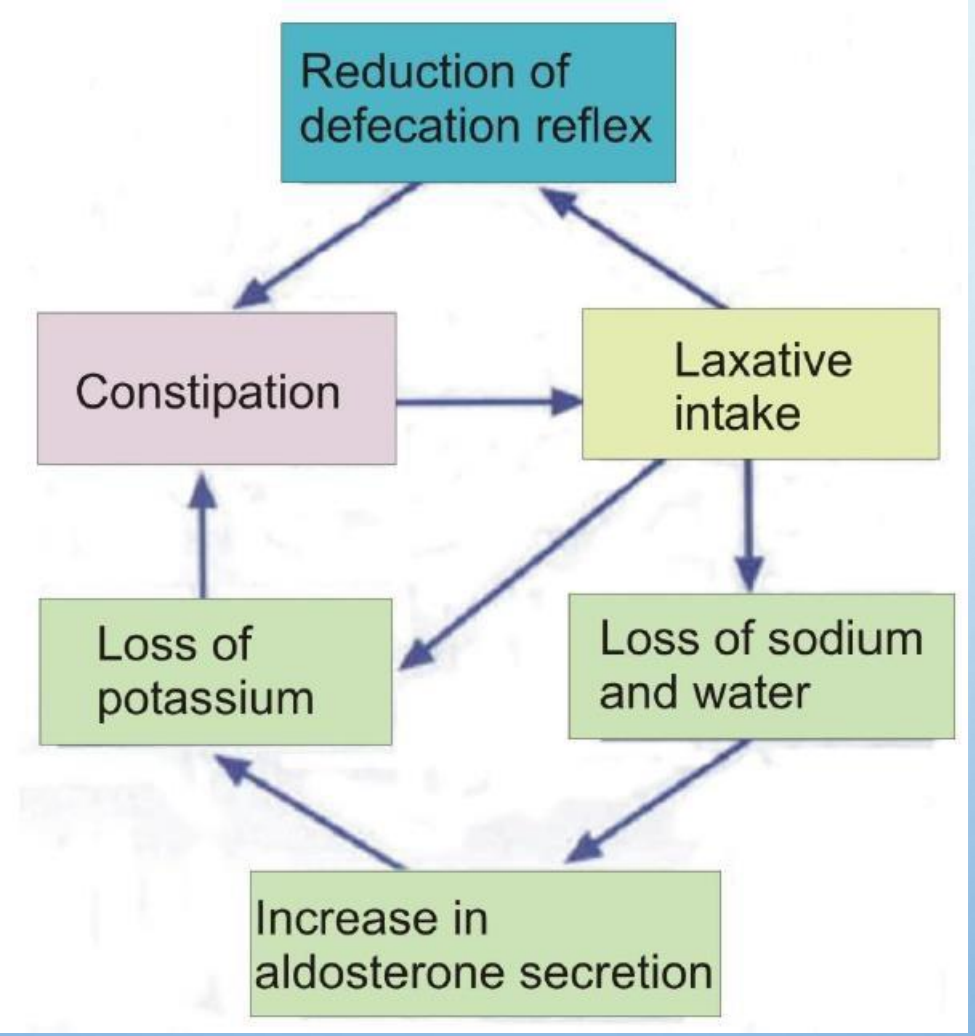
- Common in eating disorders
- Misconception:
  weight loss through use of laxatives
- Many laxatives irritate the lining of the intestines or by directly stimulating nerves.
  Continuous over-stimulation of the intestines from laxative abuse can eventually cause the bowels to become non-responsive

LAXATIVE ABUSE

Why laxatives cannot help control weight:

- Stimulate nerves in the large bowel:
  when food arrives there, nothing is left but indigestible fiber
  and non-nutritive material
- Removes water from colon as well as food residue
  → false weight loss: returns as body rehydrates after liquid intake
- Cannot stimulate small intestine!
  (This is the part of GIT where food is digested and where
  nutrients and calories are absorbed)

VICIOUS CYCLE OF CHRONIC LAXATIVE USE



Modified from Mutschler, Arzneimittelwirkung, 8th ed.

SIDE-EFFECTS OF CHRONIC LAXATIVE ABUSE

- Severe abdominal pain
- Chronic diarrhea after repeated laxative use: eventually lost control of rectum bloating
- Dehydration
- Gas
- Nausea & vomitting
- Electrolyte disturbances can lead to heart arhythmias and attacks
- Chronic constipation
- Irritable bowel syndrome
- Malabsorption of nutrients leading to hypoproteinemia, hypoalbuminemia and calcium deficiency

CONSTIPATION TREATMENT

Indications for laxative prescription:

- Constipation side-effects due to drug treatment
- Complete colonic cleaning prior to GIT endoscopic procedures
- Treatment of acute constipation

Overwhelming majority of people do not need laxatives!

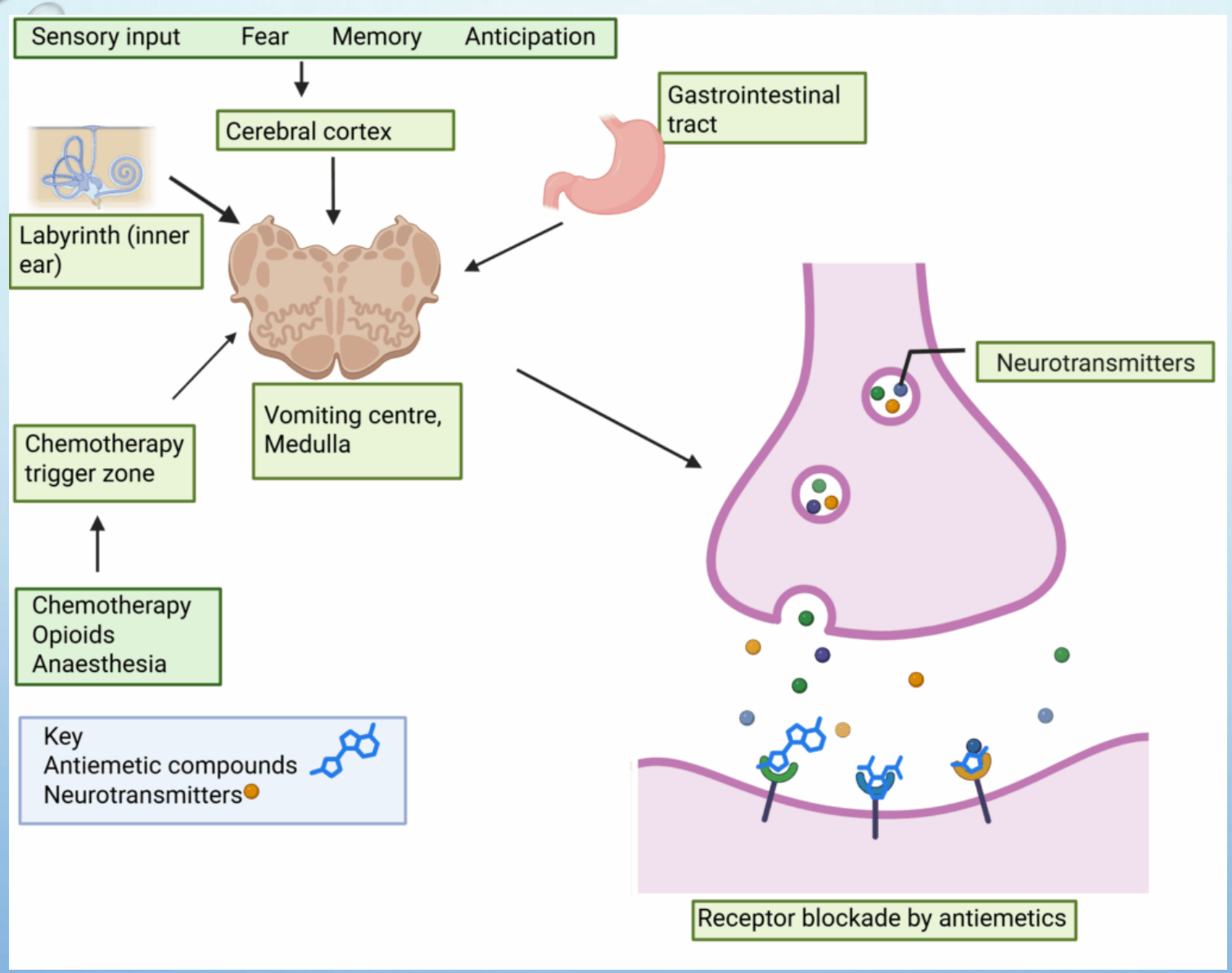
CHRONIC CONSTIPATION

Instead of laxative long-term use: → **Change of life and eating habits**

- Get enough fiber, diet with high-fiber items
  (whole grains, fresh fruits & vegetables)
- Drink plenty of water and other liquids such as fruit and vegetable juices or clear soups
- Get enough exercise
- Allow yourself enough time to have a bowel movement
- Check with your doctor about any medicine you take
  → possible side-effects

ANTI-EMETICS

- Drugs to prevent or treat nausea and vomiting



https://www.teachmepharmacy.info/gastrointestinal/drug-classes/antiemetics/

CAUSES OF NAUSEA AND VOMITING

Cause of nausea/vomiting	Preferred antiemetic classes
Motion sickness	H_1 antihistamines, Scopolamine
Gastroparesis	Metoclopramide, Domperidone
Chemotherapy	5-HT_3 antagonists, NK_1 antagonists, Dexamethasone
Postoperative nausea	5-HT_3 antagonists, D_2 antagonists
Vestibular disorders/vertigo	H_1 antihistamines, Scopolamine

ANTI-EMETIC DRUGS COMPARED

Class	Mechanism	Best uses	Major drawbacks	Example drugs
5-HT_3 antagonists	Block serotonin (5-HT_3) receptors in gut and brain	Chemotherapy, postoperative nausea, radiation-induced nausea	Constipation, headache, QT prolongation (some agents)	Ondansetron, Granisetron, Palonosetron
NK_1 antagonists	Block substance P at NK_1 receptors	Delayed chemotherapy-induced nausea	Drug interactions, higher cost	Aprepitant, Fosaprepitant
D_2 antagonists	Block dopamine receptors in CTZ	General nausea, migraine, gastroparesis	Extrapyramidal symptoms, sedation	Metoclopramide, Domperidone, Prochlorperazine
H_1 antihistamines	Block histamine H_1 receptors	Motion sickness, vertigo	Drowsiness, anticholinergic effects	Meclizine, Dimenhydrinate, Cyclizine
Antimuscarinics	Block muscarinic receptors	Motion sickness prevention	Dry mouth, blurred vision	Scopolamine (Hyoscine)
Corticosteroids	Enhances antiemetic activity via multiple mechanisms	Chemotherapy regimens, postoperative nausea	Hyperglycemia, insomnia with repeated use	Dexamethasone
Cannabinoids	Activate cannabinoid receptors	Refractory chemotherapy-related nausea	Dizziness, sedation, psychiatric effects	Dronabinol, Nabilone

The Apfel score predicts risk of postoperative nausea and vomiting based on the following risk factors

1. Sex: Female (+1)
2. Smoking: Non- smoker (+1)
3. History of motion sickness or PONV: Yes (+1)
4. Use of opioids postoperatively: Yes (+1)

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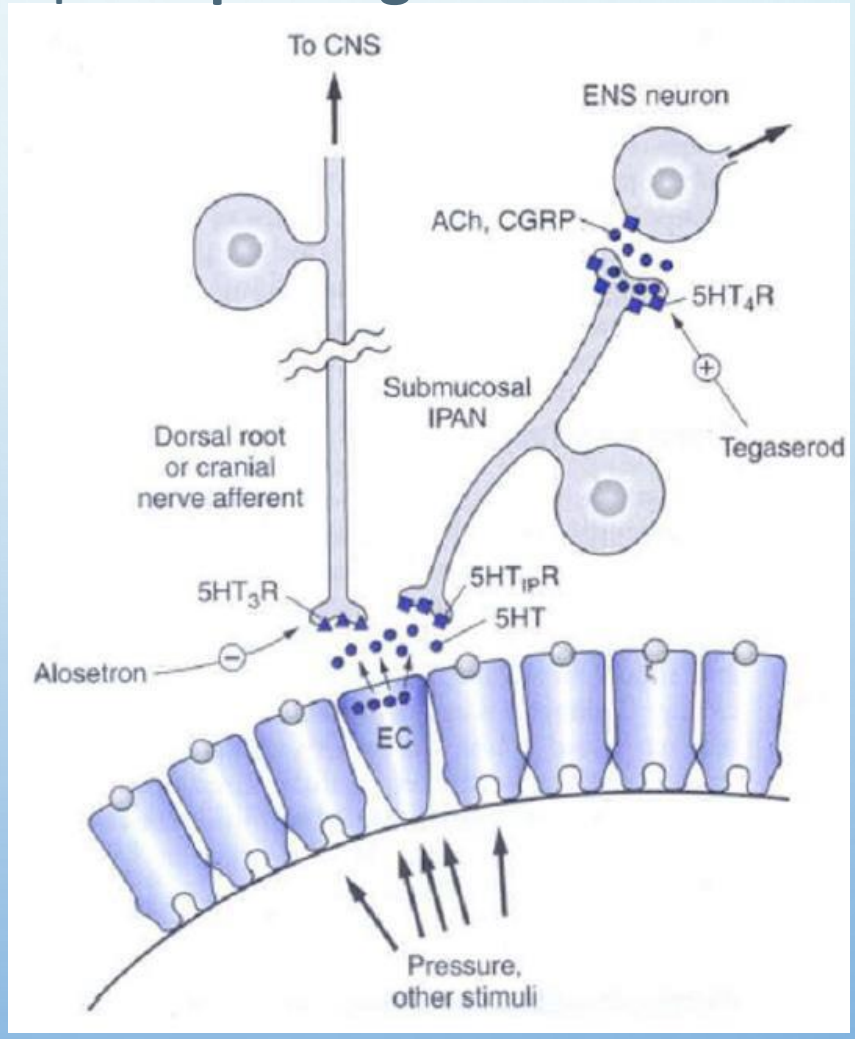
Questions? Comments?

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Serotonin 5-HT_4-receptor agonist

IPAN: intrinsic primary afferent neuron



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